

Lean Manufacturing

Engineering Technology

May, 2026

Short Title:	Lean Manufacturing	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Engineering	
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Credits	10	Level: Introductory

Description of Module / Aims

The aim of this module is to provide the student with an understanding of the Principles of Lean Manufacturing as well as gaining an important insight into its effective implementation using the fundamental tools that have played a vital role in its success.

Programmes

Higher Certificate in Engineering in Manufacturing Engineering (WD_EMANF_C) 2 / 3 / M

Indicative Content

- Fundamentals of Lean, Lean Thinking, Background, Seven Wastes, 5 Principles of Lean, Lean Tools, Continuous Improvement, JIT, Six Sigma, Education Training and Development, Teamwork.
- Standardised Work, Standard Work Chart, Standard Work Combination Chart, Problem Countermeasure Sheet.
- 5S Workplace Organisation, Sort Out, Storage, Shine, Standardise and Sustain, Numbers Game.
- Problem Solving, Problem statement, Root Cause, Direct Causes, Contributing Causes, SQC Tools, Pareto analysis, Why-Why analysis, Kepner Tregoe, SIPOC, Brainstorming, Fishbone diagrams, Corrective and Preventative action, intro to DMAIC model, Process Analytical Technology (PAT), Why-Why analysis, Polar Diagrams, Process Mapping.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the general principles, approaches, and key components of Lean and why Lean makes good business sense.
2. List the main elements of Standardised Work, understand and list the main benefits to the company and the individual.
3. Apply World Class Manufacturing (WCM) techniques to effect continuous improvement.
4. List the attributes and requirements of a well organised storage system including 5S implementation.
5. Create an accurate problem statement and perform root cause analysis, distinguishing between Direct, Contributing and Root causes.
6. Use basic tools including Pareto analysis, Brainstorming and Fishbone diagrams and distinguish between Corrective and Preventative actions.

Learning and Teaching Methods

- Lectures
- Case Studies

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3, 4, 5, 6
Continuous Assessment	50%	1,2,4,5,6
<i>Group Project</i>	15%	5,6
<i>Assignment</i>	15%	4
<i>Assignment</i>	20%	1,2

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Independent Learning	87	

Essential Material

- Bicheno, J. and M. Holweg. *_The Lean Toolbox: A handbook for lean transformation_*. 5th Ed.. .: Picsie Books, 2016.
- Liker, J. and G.L. Convis. *_The Toyota Way to Lean Leadership: Achieving and Sustaining Excellence through Leadership Development_*. 1st Ed.. .: McGraw-Hill Education, 2011.
- Wilson, L. *_How to Implement Lean Manufacturing_*. 2nd Ed.. .: McGraw-Hill Education, 2015.